setup

overhead

tag

```
(* subdirectory with nb *)
home = "projects/least squares/fireball/taylor/";
(* common variable definitions *)
Get["utility modules.m", Path → dirPack];
(* Get["Bevington modules.m",Path→dirPack]; *)
(* Get["LaTeX-tools.m",Path→dirPack]; *)
(* time, date, system user *)
stamp1;
```

```
maximum memory: 0.215694 GB
seed file: /Users/dantopa/Mathematica_files/nb/seed 23_01.nb
user: dantopa, CPU: Quaxolotl,  MM v. 13.1.0 for Mac OS X x86
date: Jan 23, 2023, time: 19:59:15
nb: /Users/dantopa/Mathematica_files/nb/projects/least squares/strays/
 seed 23-01 least squares.nb
```

modules, functions, settings, ...

source

The formation of a blast wave by a very intense explosion

I. Theoretical discussion

By Sir Geoffrey Taylor, F.R.S.

(*Received* 6 *October* 1949)

An ideal problem is here discussed. A finite amount of energy is suddenly released in an infinitely concentrated form. The motion and pressure of the surrounding air is calculated. It is found that a spherical shock wave is propagated outwards whose radius R is related to the time t since the explosion started by the equation

$$R = S(\gamma)\, t^{\frac{2}{5}} E^{\frac{1}{5}} \rho_0^{-\frac{1}{5}},$$

where ρ_0 is the atmospheric density, E is the energy released and $S(\gamma)$ a calculated function of γ, the ratio of the specific heats of air.

TABLE 1. RADIUS R OF BLAST WAVE AT TIME t AFTER THE EXPLOSION

authority	t (msec.)	R (m.)	$\log_{10} t$	$\log_{10} R$	$\frac{5}{2}\log_{10} R$
strip of small images MDDC 221	0·10	11·1	$\bar{4}$·0	3·045	7·613
	0·24	19·9	$\bar{4}$·380	3·298	8·244
	0·38	25·4	$\bar{4}$·580	3·405	8·512
	0·52	28·8	$\bar{4}$·716	3·458	8·646
	0·66	31·9	$\bar{4}$·820	3·504	8·759
	0·80	34·2	$\bar{4}$·903	3·535	8·836
	0·94	36·3	$\bar{4}$·973	3·560	8·901
strip of declassified photographs lent by Ministry of Supply	1·08	38·9	$\bar{3}$·033	3·590	8·976
	1·22	41·0	$\bar{3}$·086	3·613	9·032
	1·36	42·8	$\bar{3}$·134	3·631	9·079
	1·50	44·4	$\bar{3}$·176	3·647	9·119
	1·65	46·0	$\bar{3}$·217	3·663	9·157
	1·79	46·9	$\bar{3}$·257	3·672	9·179
	1·93	48·7	$\bar{3}$·286	3·688	9·220
strip of small images from MDDC 221	3·26	59·0	$\bar{3}$·513	3·771	9·427
	3·53	61·1	$\bar{3}$·548	3·786	9·466
	3·80	62·9	$\bar{3}$·580	3·798	9·496
	4·07	64·3	$\bar{3}$·610	3·809	9·521
	4·34	65·6	$\bar{3}$·637	3·817	9·543
	4·61	67·3	$\bar{3}$·688	3·828	9·570
large single photographs MDDC 221	15·0	106·5	$\bar{2}$·176	4·027	10·068
	25·0	130·0	$\bar{2}$·398	4·114	10·285
	34·0	145·0	$\bar{2}$·531	4·161	10·403
	53·0	175·0	$\bar{2}$·724	4·243	10·607
	62·0	185·0	$\bar{2}$·792	4·267	10·668

To compare these measurements with the analysis given in part I of this paper, equation (38) was used. It will be seen that if the ball of fire grows in the way contemplated in my theoretical analysis, $R^{\frac{5}{2}}$ will be found to be proportional to t. To find out how far this prediction was verified, the logarithmic plot of $\frac{5}{2}\log R$ against $\log t$ shown in figure 1 was made. The values from which the points were plotted are given in table 1. It will be seen that the points lie close to the 45° line which is drawn in figure 1. This line represents the relation

$$\tfrac{5}{2}\log_{10} R - \log_{10} t = 11{\cdot}915. \tag{1}$$

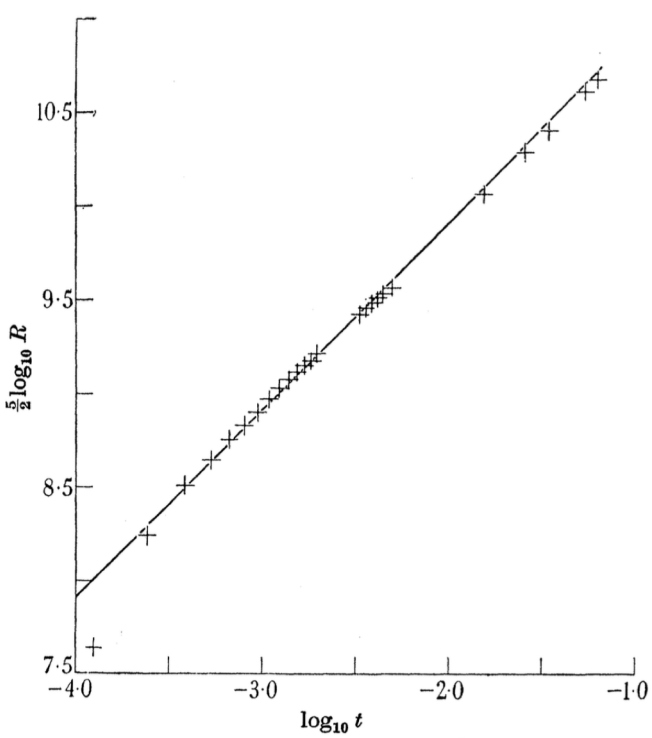

FIGURE 1. Logarithmic plot showing that $R^{\frac{5}{2}}$ is proportional to t.

Taylor's data

groups

aggregate

non-linear least squares

```
(* fireball radius in cm, time in s *)
```

In[1032]:=

```
R = {1110, 1990, 2540, 2880, 3190, 3420, 3630, 3890, 4100, 4280, 4440, 4600, 4690, 4870,
   5900, 6110, 6290, 6430, 6560, 6730, 10 650, 13 000, 14 500, 17 500, 18 500};
```

In[1033]:=

```
t = {0.0001, 0.00024, 0.00038, 0.00052, 0.00066, 0.0008, 0.00094, 0.00108,
   0.00122, 0.00136, 0.0015, 0.00165, 0.00179, 0.00193, 0.00326, 0.00353,
   0.0038, 0.00407, 0.00434, 0.00461, 0.015, 0.025, 0.034, 0.053, 0.062};
```

In[1034]:=

```
(* least squares solution *)
```

In[1035]:=

$$\text{FindMinimum}\left[\text{Norm}\left[a\, t^{b} - R, 2\right], \{a, b\}\right]$$

Out[1035]=

```
{510.098, {a → 54 473.9, b → 0.388301}}
```

end